SAEON User Stories and Tasks

The ESP32 WROOM-32 reads the TR-525M rain gauge. DHT11 is used for testing; SHT31 is the temperature and humidity sensor selected for the final product. Firebase is the database platform.

These stories define the required behaviour. They do not prove that a feature has passed testing. Final field acceptance remains open. Final-sensor evidence is tracked under PB17; Firebase access checks under PB18; portfolio completion under PB19.

US01 Monitor Rainfall

Backlog links: PB01, PB02, PB10

As a SAEON researcher, I want the station to measure rainfall automatically so that I can collect observations without manually retrieving a USB data logger.

Acceptance criteria

- The TR-525M rain gauge produces a recorded count for each bucket tip.
- Contact bounce does not produce duplicate counts during controlled testing.
- The system converts counts into rainfall depth in millimetres using the verified calibration factor for the installed gauge.

Tasks

- Connect the rain gauge to the ESP32.
- Implement tip counting and debouncing.
- Implement rainfall calculations.
- Compare recorded results with controlled test measurements.

US02 Monitor Temperature and Humidity

Backlog links: PB01, PB02, PB10, PB17

As a SAEON researcher, I want temperature and humidity readings alongside rainfall data so that I can understand conditions at the station.

Acceptance criteria

- DHT11 is identified as the testing sensor and SHT31 as the final-product sensor in configuration and test records.
- The final SHT31 output includes temperature in degrees Celsius and relative humidity in percent.
- Temperature and humidity are checked against suitable references, with the comparison method and results recorded.
- Missing or invalid readings are flagged and are not presented as valid measurements.

Tasks

- Test DHT11 readings and error handling.
- Integrate the SHT31 configuration for the final product and record the actual verification results.
- Include temperature, humidity and sensor identity in the telemetry schema and test documentation.

- Compare readings with references and record the sensor used.

US03 Transmit Data Remotely

Backlog links: PB01, PB03, PB04, PB10

As a SAEON researcher, I want the station to transmit readings through a cellular network so that I can monitor remote sites without relying on local Wi-Fi.

Acceptance criteria

- The SIM7000G establishes a data connection using the configured SIM and network settings.
- Messages reach HiveMQ Cloud over cellular MQTT and the backend ingestion service validates and stores them in Firebase.
- Each message includes a station identifier, observation time and documented measurement fields.
- Connection failures are recorded and recovery behaviour is tested.

Tasks

- Configure SIM7000G wiring, power and cellular access.
- Configure MQTT topics and access permissions.
- Configure the backend ingestion service and its Firebase writes.
- Test publishing, reconnection and stored message consistency.

US04 Store Monitoring Records

Backlog links: PB01, PB03, PB04, PB18

As a SAEON researcher, I want readings saved in a database so that I can review environmental observations over time.

Acceptance criteria

- Valid incoming readings are stored in Firebase.
- Each reading is associated with the correct station and observation time.
- Invalid messages are rejected or recorded as errors.
- Stored records can be retrieved by the dashboard.

Tasks

- Define station and telemetry records in the chosen Firebase database service.
- Develop the MQTT-to-Firebase ingestion connection.
- Validate incoming messages and enforce authorised writes.
- Test storage and retrieval through the dashboard.

US05 View Current Conditions

Backlog links: PB05

As a SAEON researcher, I want a dashboard showing the latest rainfall, temperature and humidity readings so that I can check current conditions remotely.

Acceptance criteria

- The dashboard displays rainfall, temperature and humidity with their units.
- The station identifier and latest update time are visible.
- New readings update the information displayed.
- Missing or outdated data is clearly identified.

Tasks

- Create the dashboard layout.
- Connect the dashboard to monitoring data.
- Implement live updates.
- Add loading, error, and stale-data displays.

US06 Review Historical Data

Backlog links: PB05

As a SAEON researcher, I want to filter historical readings by station and date so that I can examine changes and compare observations.

Acceptance criteria

- Users can select a station and date range.
- Charts and tables show records matching the selected filters.
- Rainfall, temperature and humidity are labelled with the correct units.
- An empty result is clearly explained.

Tasks

- Implement historical data queries.
- Add station and date filters.
- Create rainfall, temperature and humidity charts.
- Test filters against stored records.

US07 Export Monitoring Data

Backlog links: PB07

As a SAEON researcher, I want to export selected readings as a CSV file so that I can analyse them using other tools.

Acceptance criteria

- The export matches the selected station and date range.
- Column headings identify measurements and units.
- Exported values match the records displayed in the dashboard.
- Timestamps use a documented time zone.

Tasks

- Create the CSV export function.
- Apply the dashboard filters to exported data.
- Add clear column headings.
- Compare an export with database records.

US08 Check Station Health

Backlog links: PB07

As a station operator, I want to view station health information so that I can identify power or communication problems.

Acceptance criteria

- The dashboard shows the latest available battery reading and last communication time.
- Missing health information is clearly identified.
- Low battery and stale communication are flagged using agreed thresholds.
- Old readings are not presented as evidence that the station is currently healthy.

Tasks

- Implement battery measurement and calibration.
- Include diagnostic fields in transmitted messages.
- Create the station health panel.
- Test low-power and connection-loss displays.

US09 Access the Dashboard Securely

Backlog links: PB03, PB05, PB18

As an authorised SAEON user, I want to sign in securely so that monitoring information and restricted functions are available according to my access permissions.

Acceptance criteria

- Valid users can sign in and sign out.
- Restricted pages require an authenticated session.
- Database access rules enforce the agreed permissions.
- Privileged database credentials are not exposed in the browser.

Tasks

- Configure Firebase Authentication and the agreed database access rules.
- Create the sign-in interface.
- Implement protected routes and session handling.
- Test authorised access and rejected unauthorised requests.

US10 Use the Dashboard on Different Devices

Backlog links: PB07

As a SAEON researcher working away from my desk, I want a dashboard that works on phones, tablets, and computers so that I can check station information wherever I am.

Acceptance criteria

- Key readings and controls remain readable on the tested screen sizes.
- Charts, filters, and navigation remain usable.
- Important information is not hidden by overlapping or clipped elements.

Tasks

- Implement responsive dashboard layouts.
- Adapt charts and tables for smaller screens.
- Test phone, tablet, and desktop layouts.
- Correct usability issues found during testing.

US11 Maintain Station Power

Backlog links: PB02, PB06, PB09, PB10

As a station operator, I want a suitable battery and charging arrangement so that the station can continue monitoring between maintenance visits.

Acceptance criteria

- Battery, charging, and solar components are checked for compatibility.
- Operating and sleep-mode power consumption are measured.
- Charging behaviour is verified.
- Operating duration is tested against the team's agreed requirement.

Tasks

- Assemble and test the power system.
- Integrate the solar charging arrangement.
- Implement and test low-power operation.
- Record power consumption and operating duration.

US12 Deploy and Validate the Station

Backlog links: PB08, PB09, PB10, PB13

As a SAEON researcher, I want the assembled station tested in its deployment environment so that I can assess whether its measurements and communication are reliable.

Acceptance criteria

- Electronics are secured in the prepared enclosure.
- Sensor placement and station installation are documented.
- A 48-hour field test records communication, power behaviour, and measurement issues.
- Rainfall, temperature and humidity readings are compared with reference measurements.
- Results, faults, and deployment limitations are documented.

Tasks

- Complete final assembly and installation.
- Perform baseline calibration.
- Start the 48-hour field test.
- Collect manual reference measurements.
- Audit the results and record required improvements.

US13 Maintain the System

Backlog links: PB01, PB11, PB12, PB13, PB19

As a future maintainer, I want clear technical documentation so that I can set up, troubleshoot, and maintain the station and dashboard.

Acceptance criteria

- Documentation covers components, wiring, configuration, and calibration.
- Firmware, cloud, database, and dashboard setup steps are included.
- Troubleshooting guidance and known limitations are recorded.
- Credentials are handled through documented configuration steps rather than published in the documentation.
- The setup guide states Firebase, DHT11 for testing and SHT31 for the final product consistently.

Tasks

- Complete wiring and power documentation.
- Write firmware, database, and dashboard setup instructions.
- Document testing and troubleshooting.
- Review the README files for clarity and completeness.